

Ernie Chang

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Research Profile

Research scientist working on **data-centric training of foundation models**: deciding what data a model should see, in what proportion, and in what order, so that a fixed compute budget buys the most capability. Currently lead data curation and pretraining for Meta's open-source on-device *MobileLLM* and *MobileMoE* series, where a rebuilt corpus and mixture delivered roughly a $5\times$ improvement in token efficiency. Built the influence-based data-mixing framework used across text, multilingual, vision and speech training. Earlier work spans multilingual and low-resource generation, efficient and on-device modelling, and production speech interfaces. Ph.D. in Computational Linguistics.

Open-Source Models (data curation and pretraining lead)

<i>MobileLLM-R1</i>	huggingface.co/facebook/MobileLLM-R1.5-950M — on-device reasoning model.
<i>MobileLLM-Pro</i>	huggingface.co/facebook/MobileLLM-Pro — 1B on-device LLM; the language backbone selected for Meta's on-device VLM.
<i>MobileMoE</i>	huggingface.co/facebook/MobileMoE-S-Base — sparse mixture-of-experts variant for on-device deployment.

Employment History

Sep 2022 – Present	Staff Research Scientist, Meta AI ▶ Lead data curation and pretraining for the open-source on-device series <i>MobileLLM-R1</i>, <i>MobileLLM-Pro</i> and <i>MobileMoE</i>. ◦ Designed large-scale data curation and data-modelling pipelines. ◦ Proposed automatic data-mixing techniques for efficient pretraining. ◦ Developed synthetic data generation strategies for small language models. ▶ Member of the GenAI Llama team (Data and Knowledge Foundation). ◦ Data modelling for large-scale generative models: data-mixture optimisation and data-influence estimation. ◦ Led initiatives to streamline pre- and post-training data discovery and curation. ◦ Extended data-modelling techniques to multimodal settings — text, speech and video. ▶ Built the influence-based scalable data-mixing framework applied across text, multilingual, vision and speech training. ▶ Agentic post-training data: generation and scaling of synthetic reasoning and tool-calling trajectories, within a workstream where failure-mode analysis on tool-calling benchmarks (BFCL) and semantic-level quality filtering, beyond format-level checks, drive the training-data strategy. ▶ Parameter- and data-efficient modelling for speech and NLP; generative modelling for LLMs and text-to-image / text-to-music generation. ▶ Built on-device neural machine translation systems using streaming transformers.
Jul–Nov 2021 Jan–May 2022	Research Intern, Meta (Facebook) ▶ Efficient NLP methods for large-scale language models.
May 2019 – Aug 2022	Doctoral Researcher, Saarland University Department of Language Science and Technology ▶ Weak supervision for NLP, human-in-the-loop learning, and multilingual dialogue systems.
Oct 2018 – May 2019	NLP Research Engineer, Aaron.ai ▶ Built speech-based AI services: dialogue intent detection and wake-up-word (WUW) detection.
Feb – Oct 2018	ML/NLP Software Engineer, German Autolabs ▶ Implemented spoken dialogue system components: dialogue management and spoken language understanding (SLU) for an in-car voice assistant.

Apr – **R&D Intern**, eBay & Fondazione Bruno Kessler (FBK)
Aug 2017 ► Neural decoding and re-scoring framework for automated product title generation.

Education

2019 – 2022 **Ph.D., Computational Linguistics** — Saarland University, Saarbrücken
2015 – 2018 **M.S., Computational Linguistics** — University of Washington, Seattle
Thesis: *A Generative-Discriminative Framework for Title Generation in the E-commerce Domain*
2009 – 2014 **B.A., Computer Science** — University of California, San Diego

Awards

2020 **Best System Paper Award** — COLING 2020, for *DART: A Lightweight Quality-Suggestive Data-to-Text Annotation Tool*
2019 **3rd place** — Multilingual Surface Realization shared task, Deep track
2017 **6th place** — SemEval Task 1, Semantic Textual Similarity

Technical Skills

Languages Python • C++ • Java
Frameworks PyTorch • TensorFlow • MXNet • scikit-learn
Areas Data curation and mixture optimisation • data-influence estimation • synthetic data generation • large-scale pretraining and midtraining • knowledge distillation • multilingual and low-resource NLP • multimodal (vision, speech) data • on-device and streaming models

References

Yangyang Shi
Senior Staff Research Scientist
Meta AI

Vikas Chandra
Senior Director
Meta AI

Patrick Huber
Staff Research Scientist
Meta AI